

Proposal: stochastic population dynamics for STREAM

Abstract

The current STREAM model learns a deterministic, autonomous expression-space vector field from optimal-transport-paired developmental snapshots. Across balanced, partial, and unbalanced transport; PCA, KNN, and metacell denoising; adjacent and skipped intervals; and contiguous held-out blocks, causal endpoint prediction remains close to expression persistence. This proposal replaces the deterministic ordinary differential equation with a neural stochastic differential equation and trains predicted populations directly against observed endpoint populations. The drift remains a per-gene, regulatory-sequence-conditioned velocity. A low-rank diffusion term represents branching and unresolved fate variation. The recommended objective is a reparameterized, simulation-based Sinkhorn divergence that does not require individual cross-time cell pairs. We define the model, objective, regularization, evaluation, implementation stages, and decision criteria for a focused stochastic-flow ablation.

1 Motivation

The deterministic autonomous model has the form

$$\frac{dx}{dt} = v_{\theta}(E(x), \{Z_g\}_{g=1}^G), \quad (1)$$

where $x \in \mathbb{R}_{\geq 0}^G$ is expression, E is the frozen UCE encoder, and Z_g contains the AlphaGenome-derived regulatory tokens linked to gene g . The model predicts one expression velocity per gene. Training uses conditional flow-matching targets formed from optimal-transport pairs between cross-sectional timepoints.

The completed ablations do not establish meaningful improvement over persistence. On the original two-stage holdout, the best mean persistence-normalized Sinkhorn skill is approximately 0.009, while gene-level mean-shift R^2 remains negative. Holding out the middle three stages together increases the best mean Sinkhorn skill to approximately 0.02, but mean-shift R^2 is still negative. A long observed-endpoint chord across the held-out block improves some mean-shift errors without producing a consistent distributional improvement.

These results suggest that the main limitation is not simply model capacity. Cross-sectional OT does not identify a unique descendant for each source cell, development branches, and cell numbers change through proliferation and death. A deterministic map can only assign one local velocity to a state, whereas the data may support a distribution of descendants. Moreover, pairwise velocity regression is not the same objective as accurate endpoint population prediction.

2 Objective and scope

The primary question is whether representing conditional fate variability and training against endpoint distributions improves causal forecasting over persistence. The first stochastic model should therefore:

- retain gene-valued drift and the existing UCE/cCRE conditioning;

- remain autonomous, with no explicit developmental-time input;
- avoid requiring one-to-one cross-time cell pairs in the primary loss;
- generate multiple descendants per source cell;
- optimize the same type of endpoint discrepancy used for evaluation;
- separate stochastic biological variation from unrestricted technical noise through a constrained diffusion parameterization.

The initial experiment will not attempt to model division events explicitly, infer lineage trees, or reproduce raw count sampling. A growth-weighted extension is described as a second stage.

3 Autonomous neural SDE

We propose the expression-space stochastic differential equation

$$dX_t = \mu_\theta(X_t) dt + G_\phi(X_t) dW_t, \quad X_t \in \mathbb{R}_{\geq 0}^G, \quad (2)$$

where W_t is an r -dimensional Wiener process. The drift is the existing regulatory model,

$$\mu_{\theta,g}(x) = \text{STREAM}_{\theta,g}(E(x), Z_g). \quad (3)$$

Thus stochastic training does not change the interpretation of the predicted gene velocity.

3.1 Low-rank diffusion

A dense $G \times G$ diffusion matrix is neither identifiable nor tractable for $G = 10,000$. Fully independent gene noise is computationally simple but fails to represent coordinated lineage changes. We instead propose a low-rank diffusion with $r = 8\text{--}32$:

$$G_\phi(x) = \begin{bmatrix} b_{\phi,1}(E(x), Z_1)^\top \\ \vdots \\ b_{\phi,G}(E(x), Z_G)^\top \end{bmatrix} \in \mathbb{R}^{G \times r}. \quad (4)$$

Each gene’s promoter-token representation receives a shared r -dimensional diffusion head. This preserves the shared regulatory architecture and allows one latent noise direction to alter a coordinated set of genes. The diffusion head can use a bounded output,

$$b_{\phi,g} = b_{\max} \tanh(\tilde{b}_{\phi,g}), \quad (5)$$

which prevents the model from improving a distributional loss by adding unbounded noise.

The lowest-risk first implementation may instead use a state-conditioned scalar $a_\phi(E(x))$ multiplying a learned global $G \times r$ loading matrix. This is less expressive but provides a useful test of whether any coordinated stochasticity helps before adding a second regulatory output head.

4 Differentiable stochastic rollout

For an interval of length $T = t_1 - t_0$, Euler–Maruyama with K steps gives

$$X_{k+1}^{(m)} = \Pi_+ \left[X_k^{(m)} + \mu_\theta(X_k^{(m)}) \Delta t + G_\phi(X_k^{(m)}) \sqrt{\Delta t} \epsilon_k^{(m)} \right], \quad (6)$$

$$\epsilon_k^{(m)} \sim \mathcal{N}(0, I_r), \quad \Delta t = T/K, \quad (7)$$

where m indexes stochastic particles and Π_+ clamps negative expression values to zero. The Gaussian draws are external random inputs, so Equation 7 is reparameterized and gradients propagate to drift and diffusion parameters.

Online UCE preprocessing includes discrete gene sampling and is frozen. Consequently, the current encoder should be treated as a stop-gradient state map at each step. Gradients still reach earlier updates through the additive identity path in Equation 7, and each step contributes direct parameter gradients. This avoids claiming differentiability through discrete UCE tokenization. Fixed random seeds should be used within each optimization step to reduce gradient variance.

5 Recommended population-level objective

Let $\{x_i^0\}_{i=1}^B$ be source cells and $\{x_j^1\}_{j=1}^{B_1}$ an independently sampled target population. Generate M particles from every source cell and collect the predicted endpoint set

$$\hat{\mathcal{X}}_1 = \left\{ X_T^{(i,m)} : i = 1, \dots, B, m = 1, \dots, M \right\}. \quad (8)$$

The principal loss is a debiased Sinkhorn divergence in a train-only, log-normalized PCA scoring space P :

$$\mathcal{L}_{\text{pop}} = S_\varepsilon \left(P(\hat{\mathcal{X}}_1), P(\mathcal{X}_1) \right). \quad (9)$$

PCA is used here as a differentiable population metric and denoising transform, not as the model’s dynamical output space. Drift and simulated states remain gene-valued.

This loss aligns training with held-out endpoint evaluation and removes the need to select an individual OT partner for every cell. A minibatch size of $B = 128\text{--}256$, $M = 4$ particles, $K = 4$ rollout steps, and rank $r = 8$ is an appropriate initial configuration. Larger M , K , and r should be tested only after establishing a positive endpoint signal.

5.1 Regularization

Without constraints, diffusion can absorb model error or produce an overly broad population. We propose

$$\mathcal{L} = \mathcal{L}_{\text{pop}} + \lambda_G \mathbb{E} [\|G_\phi(X)\|_F^2] + \lambda_\mu \mathbb{E} [\|\mu_\theta(X)\|_2^2] + \lambda_{\text{mom}} \mathcal{L}_{\text{mom}}, \quad (10)$$

where $\mathcal{L}_{\text{mom}}$ penalizes large errors in endpoint gene means and variances. The moment term is important because the completed deterministic experiments occasionally improve Sinkhorn distance while predicting incorrect gene-level mean shifts. Diffusion should be initialized near zero, and b_{max} should be calibrated against within-timepoint variation.

To keep the distributional loss informative, target and source samples should be drawn independently at every optimization step. Validation must use fixed cell splits, seeds, source cells, target cells, and Brownian increments.

6 Cheaper paired-transition baseline

Before full rollout training, the existing OT pair banks can support a heteroscedastic transition likelihood:

$$x_1 - x_0 \sim \mathcal{N}(\mu_\theta(x_0)\Delta t, \Sigma_\phi(x_0)\Delta t). \quad (11)$$

For diagonal Σ_ϕ , the negative log likelihood is inexpensive and tests whether state-dependent uncertainty improves calibration. However, this objective inherits arbitrary OT pairing and can reward inflated variance. It should be treated as an implementation smoke test and baseline, not the main stochastic-flow result.

7 Alternatives not recommended for the first test

Noisy conditional flow matching. One can sample noisy Brownian-bridge interpolants around OT pairs and regress a time-dependent bridge drift. This is closely related to stochastic interpolants and simulation-free Schrodinger bridges [5, 6]. It requires the interpolation coordinate as an explicit model input and continues to rely on cross-time couplings. It is therefore less direct for the current autonomous, no-time-conditioning question.

Iterative Schrodinger bridge fitting. Iterative proportional fitting can learn forward and backward stochastic bridges between endpoint distributions [4]. This is a principled distributional approach but requires repeated simulation and time-dependent score or drift models. It should be considered if the simpler autonomous SDE shows a signal.

Independent per-gene noise. A diagonal diffusion head is easy to implement but mostly represents technical noise and cannot express correlated fate decisions. It remains a useful negative control against low-rank diffusion.

8 Growth and changing population mass

An SDE transports probability mass but does not create or remove it. If proliferation and death are central, add a growth-rate head $\gamma_\psi(E(x))$ and particle weights

$$w_{k+1} = w_k \exp(\gamma_\psi(X_k)\Delta t). \quad (12)$$

Training then uses weighted or unbalanced Sinkhorn divergence. This separates state motion, stochastic fate dispersion, and mass change. Growth should be introduced only after comparing deterministic and stochastic transport under fixed mass; otherwise diffusion and growth may compensate for each other and be difficult to identify.

9 Evaluation

For each source cell, generate $M_{\text{eval}} = 16\text{--}64$ particles with fixed seeds. Report:

- debiased Sinkhorn divergence and skill relative to persistence;
- gene-level mean-shift R^2 and mean absolute error;
- mean per-gene Wasserstein-1 distance;
- endpoint covariance error in train-only PCA space;
- coverage and calibration of observed cells under generated particles;
- metrics stratified by annotated cell type or lineage where available;
- diffusion-disabled rollout from the same checkpoint;
- sensitivity to particle count and integration steps.

Persistence should be represented by repeated source cells, so it has the same number of particles as the stochastic model. A deterministic STREAM model, the same SDE with $G = 0$, and a pure-diffusion model with $\mu = 0$ provide the necessary ablations. Pairwise displacement R^2 should remain secondary because no individual descendant is observed.

Table 1: Proposed staged experiments. Advancement requires improved held-out population metrics without loss of gene-level mean-shift accuracy.

Stage	Model	Training objective	Decision
0	Existing deterministic drift	Existing CFM loss	Reproduce current checkpoint and evaluation
1	Diagonal heteroscedastic transition	OT-paired Gaussian NLL	Verify diffusion code and calibration
2	Global rank-8 diffusion	Population Sinkhorn, $M = 4$, $K = 4$	Require positive skill and non-degraded mean-shift metrics
3	CRE-conditioned rank-8 diffusion	Population Sinkhorn plus moment loss	Test whether state/gene-specific uncertainty helps
4	Best SDE with $K = 8, 16$ and $M = 8$	Same objective	Check numerical and particle convergence
5	Optional growth head	Weighted Sinkhorn unbalanced	Proceed only if mass mismatch remains limiting

10 Experimental plan

The first full comparison should use the contiguous middle-three-stage holdout. Train separate adjacent, skip-1, and skip-2 models only if the rank-8 Stage 2 model beats persistence. The observed bridge interval across the block can be added as a subsequent controlled factor.

11 Implementation plan

The implementation should preserve package boundaries:

1. Add drift–diffusion output structures to the reusable model package. Keep the current drift-only checkpoint path backward compatible.
2. Add a reusable Euler–Maruyama rollout accepting fixed Gaussian noise, particle count, diffusion rank, and nonnegativity projection.
3. Add differentiable train-only PCA scoring and debiased Sinkhorn loss in torch. Do not route model outputs through PCA.
4. Add an interval-stratified population sampler that returns independent source and target cells rather than OT pairs.
5. Add fixed stochastic validation batches containing cell indices and Brownian seeds. Select checkpoints by endpoint Sinkhorn EMA, with moment metrics as safeguards.
6. Extend causal evaluation to generate multiple particles, report stochastic calibration, and retain deterministic/persistence baselines.
7. Run synthetic tests where a branching SDE has known drift and diffusion before fitting atlas data.

The highest-risk engineering component is repeated online UCE evaluation for $B \times M$ particles. Initial rollout training should cache repeated source encodings where possible, use small M and K , and profile whether UCE or the gene-conditioned drift dominates runtime. A lower-cost expression-state SDE can validate the training objective before enabling online UCE.

12 Success and stopping criteria

The stochastic model should be considered useful only if it satisfies all of the following on held-out stages:

- Sinkhorn skill is consistently positive across horizons and materially larger than the current one-to-two percent range;
- mean-shift R^2 is nonnegative or improves without a compensating degradation in Wasserstein distance;
- performance is stable across Brownian seeds and particle counts;
- learned diffusion is nonzero, bounded, and improves over both drift-only and pure-diffusion controls;
- improvement persists within major cell types rather than arising only from global mixture smoothing.

If population-level SDE training still matches persistence, the appropriate conclusion is that expression snapshots and the present state representation do not identify the requested dynamics at this resolution. Further architecture sweeps would then be lower priority than lineage constraints, explicit cell types, perturbations, or direct proliferation information.

References

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